

STAGE 5 INDUSTRIAL TECHNOLOGY ELECTIVE - TIMBER INDUSTRY

Timber Pen | 50-period teaching program

Website-complete teacher review copy | Industrial Technology 7-10 Syllabus (2025)

Course	Stage 5 Industrial Technology elective - Timber industry	Delivery	2027 20 weeks
Timetable	50 x 60-minute periods	Pattern	5 periods per fortnight
Lesson balance	10 theory 10 project/folio 30 practical	Syllabus	2025 syllabus - approved early use
Status	Local teacher review only	Project	Produce the approved timber pen project
Website coverage: 30 named theory sections, 10 quiz/check sets (120 questions) and 40 scaffolded responses are scheduled. Practical lessons remain the majority.			

Industrial Technology outcomes

Outcome	Official wording
INT5-SAF-01	applies risk management and safe work practices in industrial technology contexts
INT5-COM-01	communicates ideas and processes in industrial technology contexts
INT5-ENV-01	assesses the impact of industrial activities and practices on society and the environment
INT5-DES-01	selects and applies design, project management and technical skills in project production
INT5-USE-01	selects and justifies the use of materials, tools and equipment to undertake projects
INT5-EVL-01	evaluates the functional, aesthetic, economic and environmental qualities of products
INT5-INP-01	analyses and applies a range of industrial processes in production contexts
INT5-PRO-01	demonstrates practical skills in the production of projects
INT5-IND-01	investigates and explains a range of industry careers, enterprise opportunities and practices

Teacher to confirm before use

When	Confirm	Owner
Before Period 1	Assessment dates, submission method, class supports and adjustments	Teacher / faculty
Before practical work	Current plan revision, permitted variations, induction, SOPs, tools, PPE, supervision and risk controls	Teacher / faculty
Before finishing	Approved finish, SDS/product instructions, area, drying and storage arrangements	Teacher / faculty

Teaching program | Periods 1-10

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
1	Site/theory	Weeks 1-2 website	Understanding the Pen project brief and success criteria. Weeks 1-2 knowledge checks (questions 1-4). scaffolded responses 1-2	Design factors and project specifications	Website work checked INT5-DES-01, INT5-COM-01
2	Project/folio	Weeks 1-2 folio	Reading the approved Pen Making process resource and kit instructions. Weeks 1-2 knowledge checks (questions 5-8). scaffolded response 3. Folio: link to the project; set the next step.	Safe wood-turning setup and practice	Folio + next step INT5-SAF-01, INT5-INP-01
3	Practical	Weeks 1-2 site + practical	First 20 min: Managing wood-lathe hazards, risks and teacher approval. Weeks 1-2 knowledge checks (questions 9-12). scaffolded response 4. Then approved stage; check plan, setup and one quality point.	Risk management and risk-reduction procedures	Website + practical check INT5-SAF-01, INT5-PRO-01
4	Practical	Weeks 1-2 production	Continue approved stage; follow plan/SOP; check fit, accuracy or surface.	Risk management and risk-reduction procedures	Practical check INT5-SAF-01, INT5-PRO-01
5	Practical	Weeks 1-2 evidence	Finish current stage; photograph a checkpoint; record any problem/response; clean down.	Prepare timber surfaces for finishes	Photo + quality note INT5-INP-01, INT5-PRO-01
6	Site/theory	Weeks 3-4 website	Selecting and inspecting timber for the Pen blank. Weeks 3-4 knowledge checks (questions 1-4). scaffolded responses 1-2	Timber properties and justified material selection	Website work checked INT5-USE-01, INT5-ENV-01
7	Project/folio	Weeks 3-4 folio	Using grain orientation to keep the two Pen bodies visually connected. Weeks 3-4 knowledge checks (questions 5-8). scaffolded response 3. Folio: link to the project; set the next step.	Timber properties and justified material selection	Folio + next step INT5-USE-01, INT5-ENV-01
8	Practical	Weeks 3-4 site + practical	First 20 min: Planning responsible material use and the Pen cutting list. Weeks 3-4 knowledge checks (questions 9-12). scaffolded response 4. Then approved stage; check plan, setup and one quality point.	Accurate measuring and marking-out for timber projects	Website + practical check INT5-USE-01, INT5-PRO-01
9	Practical	Weeks 3-4 production	Continue approved stage; follow plan/SOP; check fit, accuracy or surface.	Risk management and risk-reduction procedures	Practical check INT5-SAF-01, INT5-PRO-01
10	Practical	Weeks 3-4 evidence	Finish current stage; photograph a checkpoint; record any problem/response; clean down.	Prepare timber surfaces for finishes	Photo + quality note INT5-INP-01, INT5-PRO-01

Teaching program | Periods 11-20

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
11	Site/theory	Weeks 5-6 website	Marking the Pen billet and preserving reliable references. Weeks 5-6 knowledge checks (questions 1-4). scaffolded responses 1-2	Accurate measuring and marking-out for timber projects	Website work checked INT5-USE-01, INT5-PRO-01
12	Project/folio	Weeks 5-6 folio	Preparing the two brass sleeves for the approved Pen kit. Weeks 5-6 knowledge checks (questions 5-8). scaffolded response 3. Folio: link to the project; set the next step.	Safe wood-turning setup and practice	Folio + next step INT5-SAF-01, INT5-INP-01
13	Practical	Weeks 5-6 site + practical	First 20 min: Drilling clean Pen blank holes and clearing swarf. Weeks 5-6 knowledge checks (questions 9-12). scaffolded response 4. Then approved stage; check plan, setup and one quality point.	Accurate measuring and marking-out for timber projects	Website + practical check INT5-USE-01, INT5-PRO-01
14	Practical	Weeks 5-6 production	Continue approved stage; follow plan/SOP; check fit, accuracy or surface.	Risk management and risk-reduction procedures	Practical check INT5-SAF-01, INT5-PRO-01
15	Practical	Weeks 5-6 evidence	Finish current stage; photograph a checkpoint; record any problem/response; clean down.	Prepare timber surfaces for finishes	Photo + quality note INT5-INP-01, INT5-PRO-01
16	Site/theory	Weeks 7-8 website	Understanding between-centres turning and the approved Pen mandrel. Weeks 7-8 knowledge checks (questions 1-4). scaffolded responses 1-2	Safe wood-turning setup and practice	Website work checked INT5-SAF-01, INT5-INP-01
17	Project/folio	Weeks 7-8 folio	Assembling the approved mandrel with three collets and brass thumb screw. Weeks 7-8 knowledge checks (questions 5-8). scaffolded response 3. Folio: link to the project; set the next step.	Document, reflect on and evaluate project development	Folio + next step INT5-COM-01, INT5-DES-01
18	Practical	Weeks 7-8 site + practical	First 20 min: Lathe pre-start checks and secure approved workholding. Weeks 7-8 knowledge checks (questions 9-12). scaffolded response 4. Then approved stage; check plan, setup and one quality point.	Accurate measuring and marking-out for timber projects	Website + practical check INT5-USE-01, INT5-PRO-01
19	Practical	Weeks 7-8 production	Continue approved stage; follow plan/SOP; check fit, accuracy or surface.	Risk management and risk-reduction procedures	Practical check INT5-SAF-01, INT5-PRO-01
20	Practical	Weeks 7-8 evidence	Finish current stage; photograph a checkpoint; record any problem/response; clean down.	Prepare timber surfaces for finishes	Photo + quality note INT5-INP-01, INT5-PRO-01

Teaching program | Periods 21-30

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
21	Site/theory	Weeks 9-10 website	From square blank to controlled cylindrical form. Weeks 9-10 knowledge checks (questions 1-4). scaffolded responses 1-2	Document, reflect on and evaluate project development	Website work checked INT5-COM-01, INT5-DES-01
22	Project/folio	Weeks 9-10 folio	Understanding cutting and scraping during Pen turning. Weeks 9-10 knowledge checks (questions 5-8). scaffolded response 3. Folio: link to the project; set the next step.	Safe wood-turning setup and practice	Folio + next step INT5-SAF-01, INT5-INP-01
23	Practical	Weeks 9-10 site + practical	First 20 min: Monitoring profile, alignment and material removal against approved kit references. Weeks 9-10 knowledge checks (questions 9-12). scaffolded response 4. Then approved stage; check plan, setup and one quality point.	Accurate measuring and marking-out for timber projects	Website + practical check INT5-USE-01, INT5-PRO-01
24	Practical	Weeks 9-10 production	Continue approved stage; follow plan/SOP; check fit, accuracy or surface.	Risk management and risk-reduction procedures	Practical check INT5-SAF-01, INT5-PRO-01
25	Practical	Weeks 9-10 evidence	Finish current stage; photograph a checkpoint; record any problem/response; clean down.	Prepare timber surfaces for finishes	Photo + quality note INT5-INP-01, INT5-PRO-01
26	Site/theory	Weeks 11-12 website	Diagnosing tool marks and surface defects before sanding. Weeks 11-12 knowledge checks (questions 1-4). scaffolded responses 1-2	Accurate measuring and marking-out for timber projects	Website work checked INT5-USE-01, INT5-PRO-01
27	Project/folio	Weeks 11-12 folio	Using a progressive approved abrasive sequence. Weeks 11-12 knowledge checks (questions 5-8). scaffolded response 3. Folio: link to the project; set the next step.	Document, reflect on and evaluate project development	Folio + next step INT5-COM-01, INT5-DES-01
28	Practical	Weeks 11-12 site + practical	First 20 min: Selecting and applying the approved Pen finish from product information. Weeks 11-12 knowledge checks (questions 9-12). scaffolded response 4. Then approved stage; check plan, setup and one quality point.	Accurate measuring and marking-out for timber projects	Website + practical check INT5-USE-01, INT5-PRO-01
29	Practical	Weeks 11-12 production	Continue approved stage; follow plan/SOP; check fit, accuracy or surface.	Risk management and risk-reduction procedures	Practical check INT5-SAF-01, INT5-PRO-01
30	Practical	Weeks 11-12 evidence	Finish current stage; photograph a checkpoint; record any problem/response; clean down.	Prepare timber surfaces for finishes	Photo + quality note INT5-INP-01, INT5-PRO-01

Teaching program | Periods 31-40

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
31	Site/theory	Weeks 13-14 website	Identifying and staging the Pen kit components for assembly. Weeks 13-14 knowledge checks (questions 1-4). scaffolded responses 1-2	Assembly using fixings, adhesives or clamps	Website work checked INT5-INP-01, INT5-PRO-01
32	Project/folio	Weeks 13-14 folio	Assembling the clip/button and nib shroud in the approved sequence. Weeks 13-14 knowledge checks (questions 5-8). scaffolded response 3. Folio: link to the project; set the next step.	Document, reflect on and evaluate project development	Folio + next step INT5-COM-01, INT5-DES-01
33	Practical	Weeks 13-14 site + practical	First 20 min: Preventing misalignment and damage during Pen assembly. Weeks 13-14 knowledge checks (questions 9-12). scaffolded response 4. Then approved stage; check plan, setup and one quality point.	Accurate measuring and marking-out for timber projects	Website + practical check INT5-USE-01, INT5-PRO-01
34	Practical	Weeks 13-14 production	Continue approved stage; follow plan/SOP; check fit, accuracy or surface.	Risk management and risk-reduction procedures	Practical check INT5-SAF-01, INT5-PRO-01
35	Practical	Weeks 13-14 evidence	Finish current stage; photograph a checkpoint; record any problem/response; clean down.	Prepare timber surfaces for finishes	Photo + quality note INT5-INP-01, INT5-PRO-01
36	Site/theory	Weeks 15-16 website	Designing a comfortable and visually resolved Pen profile within kit constraints. Weeks 15-16 knowledge checks (questions 1-4). scaffolded responses 1-2	Design factors and project specifications	Website work checked INT5-DES-01, INT5-COM-01
37	Project/folio	Weeks 15-16 folio	Standards, quality control and production techniques in Pen manufacture. Weeks 15-16 knowledge checks (questions 5-8). scaffolded response 3. Folio: link to the project; set the next step.	Safe wood-turning setup and practice	Folio + next step INT5-SAF-01, INT5-INP-01
38	Practical	Weeks 15-16 site + practical	First 20 min: Comparing handcraft, jigs and repeat production technologies. Weeks 15-16 knowledge checks (questions 9-12). scaffolded response 4. Then approved stage; check plan, setup and one quality point.	Accurate measuring and marking-out for timber projects	Website + practical check INT5-USE-01, INT5-PRO-01
39	Practical	Weeks 15-16 production	Continue approved stage; follow plan/SOP; check fit, accuracy or surface.	Risk management and risk-reduction procedures	Practical check INT5-SAF-01, INT5-PRO-01
40	Practical	Weeks 15-16 evidence	Finish current stage; photograph a checkpoint; record any problem/response; clean down.	Prepare timber surfaces for finishes	Photo + quality note INT5-INP-01, INT5-PRO-01

Teaching program | Periods 41-50

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
41	Site/theory	Weeks 17-18 website	Building a real Pen Work Method Statement. Weeks 17-18 knowledge checks (questions 1-4). scaffolded responses 1-2	Safe wood-turning setup and practice	Website work checked INT5-SAF-01, INT5-INP-01
42	Project/folio	Weeks 17-18 folio	Creating the Pen production flowchart: inputs, processes and outputs. Weeks 17-18 knowledge checks (questions 5-8). scaffolded response 3. Folio: link to the project; set the next step.	Safe wood-turning setup and practice	Folio + next step INT5-SAF-01, INT5-INP-01
43	Practical	Weeks 17-18 site + practical	First 20 min: Explaining social and environmental impacts of Pen production and use. Weeks 17-18 knowledge checks (questions 9-12). scaffolded response 4. Then approved stage; check plan, setup and one quality point.	Accurate measuring and marking-out for timber projects	Website + practical check INT5-USE-01, INT5-PRO-01
44	Practical	Weeks 17-18 production	Continue approved stage; follow plan/SOP; check fit, accuracy or surface.	Risk management and risk-reduction procedures	Practical check INT5-SAF-01, INT5-PRO-01
45	Practical	Weeks 17-18 evidence	Finish current stage; photograph a checkpoint; record any problem/response; clean down.	Prepare timber surfaces for finishes	Photo + quality note INT5-INP-01, INT5-PRO-01
46	Site/theory	Weeks 19-20 website	Photographing Pen stages and explaining problem solving. Weeks 19-20 knowledge checks (questions 1-4). scaffolded responses 1-2	Safe wood-turning setup and practice	Website work checked INT5-SAF-01, INT5-INP-01
47	Project/folio	Weeks 19-20 folio	Using PMI to evaluate the completed Pen. Weeks 19-20 knowledge checks (questions 5-8). scaffolded response 3. Folio: link to the project; set the next step.	Safe wood-turning setup and practice	Folio + next step INT5-SAF-01, INT5-INP-01
48	Practical	Weeks 19-20 site + practical	First 20 min: Final reflection, evidence review and skill transfer. Weeks 19-20 knowledge checks (questions 9-12). scaffolded response 4. Then approved stage; check plan, setup and one quality point.	Accurate measuring and marking-out for timber projects	Website + practical check INT5-USE-01, INT5-PRO-01
49	Practical	Weeks 19-20 production	Continue approved stage; follow plan/SOP; check fit, accuracy or surface.	Risk management and risk-reduction procedures	Practical check INT5-SAF-01, INT5-PRO-01
50	Practical	Weeks 19-20 evidence	Finish current stage; photograph a checkpoint; record any problem/response; clean down.	Prepare timber surfaces for finishes	Photo + quality note INT5-INP-01, INT5-PRO-01